



RTL Design Sherpa

RAPIDS Byte-Granular DMA Engine Byte Characterization Report

September 30, 2026

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Readme

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RAPIDS Byte-Granular DMA: Byte Characterization Report

Version: 0.2
Date: 2026-09-30
Platform: Genesys 2, 256-bit, 8 channels, 100 MHz
Results: rapids_byte_perf_20260930_142731.json

1. Headline

Largest transfers at 8 channels

Path	Point	Efficiency	MB/s measured	% of 3200 MB/s peak
Sink	byte path, 4096 B x 8 ch	1.000	357	11.1 %
Sink	beat path, 4096 beats x 8 ch	1.000	356	11.1 %
Sink	beat path, 4096 beats x 8 ch, word-wide checker build	1.000	3182	99.4 %
Source	byte path, 4096 B x 8 ch	1.000	352	11.0 %
Source	beat path, 4096 beats x 8 ch	1.000	356	11.1 %
Source	beat path, 4096 beats x 8 ch, word-wide	1.000	3199	100.0 %

Path	Point	Efficiency	MB/s measured	% of 3200 MB/s peak
checker build				

The byte-wise rates sit at the harness checker ceiling of $3200 / 9 = 355.6$ MB/s, not at the DUT's limit: the byte-wise CRC checkers take 9 cycles per 32-byte beat (section 3). Read every MB/s in this report against both the 3200 MB/s peak and that ceiling. Efficiency (payload over beats x lanes) does not depend on the checker.

The word-wide checker build (BYTE_CRC=0) takes one beat per cycle and is not checker-bound: its rows show the DUT's own rate against the 3200 MB/s peak. Its data check covers 4 of the 32 bytes of each beat, so it is a rate measurement and the byte-wise build is the integrity measurement.

2. Definitions

Definitions

Term	Meaning
Bytes moved	Total payload bytes over all active channels and descriptors; from the exact byte count of the AXIS bus meter.
Beats moved	Accepted beats on the interface named: AXIS beats on the stream side, memory beats on the AXI side.
Efficiency	payload bytes / (beats x BYTE_LANES), BYTE_LANES = 32. 1.0 means every lane of every beat carried payload.
MB/s	Total bytes / measurement window, windows counted in 100 MHz cycles by the harness meters, not wall clock.
Peak	$3200 \text{ MB/s} = 100 \text{ MHz} \times 32 \text{ B}$, per direction. Shown beside every measured MB/s value.
Engaged utilization	prod / (prod + bp + starv) on one interface; the RAPIDS Beats headline metric, used only for the beat-aligned comparison.

3. Beat-aligned rows against the RAPIDS Beats report

These rows use the beat-scaled path (pkt_bytes=None, beats per channel) on the byte DUT, which is how the RAPIDS Beats report measured its matrix: 9-beat bursts, response delay 0, backpressure off, default seed, bare bus meters. The metric is engaged utilization ($\text{prod} / (\text{prod} + \text{bp} + \text{starv})$), the RAPIDS Beats headline metric. “Unchanged” is checked numerically: every cell is compared to the same cell of genesys_dw256_obs_C.json and must agree within 0.5 percentage points, the threshold the RAPIDS Beats report itself applies between builds.

3.1 Word-wide checker build (the verdict)

Measured on the BYTE_CRC=0 bitstream (BUILD.WORD_CRC = 1, results file rapids_byte_aligned_wordcrc_20260930.json, bitstream sha256 48e1282c46e707cd). Its checkers take one beat per cycle like the RAPIDS Beats build. They CRC one 32-bit slice of each beat (slice 0, which holds the beat’s LFSR word) and ignore the strobes, so the golden is the RAPIDS Beats golden and the data check covers 4 of the 32 bytes of each beat. That is enough for a utilization measurement of whole-beat rows; full-byte integrity is what the byte-wise build in 3.2 and the rest of this report check. The design under test is the same RTL.

Engaged utilization, byte build / RAPIDS Beats reference, beat-aligned rows

Ch	Beats/ch	Sink AXIS- in % (byte / beats)	Sink AXI4- wr %	Source AXI4-rd %	Source AXIS-out %	Max abs delta (pp)
1	1	50.0 / 50.0	10.0 / 14.3	6.2 / 6.7	6.2 / 6.7	4.29
1	4	100.0 / 80.0	25.0 / 40.0	21.1 / 22.2	21.1 / 22.2	20.00
1	16	25.0 / 94.1	48.5 / 72.7	51.6 / 53.3	51.6 / 53.3	69.12
1	64	57.1 / 98.5	79.0 / 91.4	81.0 / 82.1	81.0 / 82.1	41.32
1	256	84.2 / 99.6	93.8 / 97.7	94.5 / 94.8	94.5 / 94.8	15.40
1	1024	95.5 / 99.9	98.4 / 99.4	98.6 / 98.7	98.6 / 98.7	4.38
1	4096	98.8 / 100.0	99.6 / 99.9	99.6 / 99.7	99.6 / 99.7	1.13
2	1	66.7 / 66.7	16.7 / 22.2	11.1 / 11.8	11.1 / 11.8	5.56
2	4	10.5 / 88.9	40.0 / 57.1	34.8 / 36.4	34.8 / 36.4	78.36
2	16	32.0 / 97.0	65.3 / 84.2	68.1 / 69.6	68.1 / 69.6	64.97
2	64	65.3 / 99.2	88.3 / 95.5	89.5 / 90.1	89.5 / 90.1	33.92
2	256	88.3 / 99.8	96.8 / 98.8	97.2 / 97.3	97.2 / 97.3	11.53
2	1024	96.8 /	99.2 / 99.7	99.3 / 99.3	99.3 / 99.3	3.16

Ch	Beats/ch	Sink AXIS- in % (byte / beats)	Sink AXI4- wr %	Source AXI4-rd %	Source AXIS-out %	Max abs delta (pp)
		100.0				
2	4096	99.2 /	99.8 / 99.9	99.8 / 99.8	99.8 / 99.8	0.81
		100.0				
4	1	100.0 /	25.0 / 30.8	18.2 / 19.0	18.2 / 19.0	20.00
		80.0				
4	4	12.9 / 94.1	57.1 / 72.7	51.6 / 53.3	51.6 / 53.3	81.21
4	16	37.2 / 98.5	79.0 / 91.4	81.0 / 82.1	81.0 / 82.1	61.25
4	64	70.3 / 99.6	93.8 / 97.3	94.5 / 94.8	94.5 / 94.8	29.28
4	256	90.5 / 99.7	98.4 / 99.4	98.6 / 98.7	98.6 / 98.7	9.25
4	1024	97.4 / 99.9	99.6 / 99.9	99.6 / 99.7	99.6 / 99.7	2.50
4	4096	99.3 /	99.9 /	99.9 / 99.9	99.9 / 99.9	0.64
		100.0	100.0			
8	1	4.0 / 88.9	33.3 / 38.1	26.7 / 27.6	26.7 / 27.6	84.85
8	4	14.5 / 97.0	72.7 / 84.2	68.1 / 69.6	68.1 / 69.6	82.42
8	16	40.5 / 99.2	88.3 / 95.5	89.5 / 90.1	89.5 / 90.1	58.72
8	64	73.1 / 99.8	96.8 / 98.5	97.2 / 97.3	97.2 / 97.3	26.66
8	256	91.6 / 96.1	99.2 / 99.7	99.3 / 99.3	99.3 / 99.3	4.51
8	1024	97.8 / 99.0	99.8 / 99.9	99.8 / 99.8	99.8 / 99.8	1.24
8	4096	99.4 / 99.7	99.9 /	100.0 /	100.0 /	0.32
			100.0	100.0	100.0	

Verdict: CHANGED. 112 of 112 beat-aligned cells were compared; 73 differ by more than 0.5 pp; the largest difference is -84.85 pp (ch 8, 1 beats, sink sin). Cells over the threshold: ch1 b1 sink wr -4.29 pp; ch1 b4 sink sin +20.00 pp; ch1 b4 sink wr -15.00 pp; ch1 b4 source rd -1.17 pp; ch1 b4 source sout -1.17 pp; ch1 b16 sink sin -69.12 pp; ch1 b16 sink wr -24.24 pp; ch1 b16 source rd -1.72 pp; ch1 b16 source sout -1.72 pp; ch1 b64 sink sin -41.32 pp; ch1 b64 sink wr -12.42 pp; ch1 b64 source rd -1.04 pp;

Where the differences come from. Every difference is a fixed number of cycles per run, not a change of rate. The cycle counts below are the evidence; the same counts at 16 beats per channel are identical to these, and only the 1-beat rows differ (the write side shows +3 starvation cycles there, not +11).

Cycle counts at 4096 beats per channel (fixed start-up terms, not rates)

Ch	Sink AXIS-in bp, byte	beats	Sink AXI4-wr starv, byte	beats	Source AXI4-rd starv, byte	beats
1	47	0	17	6	15	14
2	67	0	17	6	15	14
4	107	2	17	6	15	14
8	187	82	17	6	15	14

- **Sink AXIS-in.** The byte ingress accepts a channel's stream only once that channel has a packet record, because the destination offset and the expected length come from the descriptor (sink ingress chapter of the MAS). The stream is offered data, t ready is low, and the meter counts those cycles as backpressure. The count grows with the channel count and does not depend on the beats per channel, which is what a serial descriptor fetch before the first accept would give. RAPIDS Beats has no such term because it buffers the stream before the descriptor arrives. The AXIS-in window opens at the first offered beat, so this wait is inside it.
- **Sink AXI4-wr.** The write-side window opens at the first write, so the wait above is not in it. The byte build adds a constant 11 starvation cycles at 16 beats and up (3 at 1 beat), the same for every channel count. Its source has not been isolated.
- **Source.** One extra starvation cycle at start-up in every cell, on both rd and sout: up to 1.7 percentage points at 1 to 64 beats, under 0.4 from 256 beats. Not isolated.
- **Amortisation.** Because each term is fixed, utilization converges on the RAPIDS Beats value as the transfer grows: the 4096-beat rows are within 1.2 percentage points and the 8-channel row within 0.4. The 1-to-64-beat rows, where the fixed term is a large share of the window, are the ones that move most.
- **Not explained.** With 1 beat per channel and 1, 2 or 4 channels the byte build shows no sink AXIS-in backpressure, while the same channels at 16 beats and up do. Not isolated.

3.2 Byte-wise checker build (the standard bitstream)

Engaged utilization, byte build / RAPIDS Beats reference, beat-aligned rows

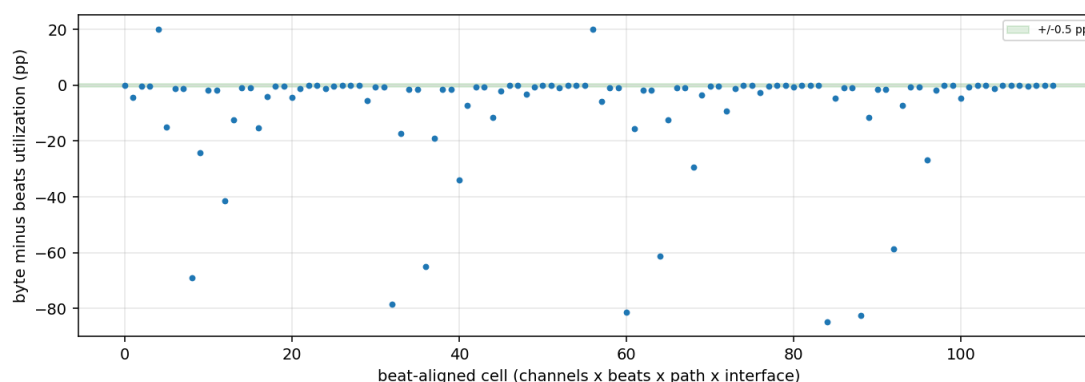
Ch	Beats/ch	Sink AXIS- in % (byte / beats)	Sink AXI4- wr %	Source AXI4-rd %	Source AXIS-out %	Max abs delta (pp)
1	1	50.0 / 50.0	10.0 / 14.3	6.2 / 6.7	6.7 / 6.7	4.29
1	4	100.0 / 80.0	25.0 / 40.0	9.3 / 22.2	9.5 / 22.2	20.00
1	16	25.0 / 94.1	13.6 / 72.7	10.6 / 53.3	10.7 / 53.3	69.12
1	64	57.1 / 98.5	11.6 / 91.4	20.7 / 82.1	10.7 / 82.1	79.79
1	256	22.6 / 99.6	11.2 / 97.7	18.6 / 94.8	11.0 / 94.8	86.47
1	1024	12.7 / 99.9	11.1 / 99.4	12.4 / 98.7	11.1 / 98.7	88.27
1	4096	11.5 / 100.0	11.1 / 99.9	11.4 / 99.7	11.1 / 99.7	88.73
2	1	66.7 / 66.7	16.7 / 22.2	8.0 / 11.8	8.7 / 11.8	5.56
2	4	10.5 / 88.9	19.5 / 57.1	10.1 / 36.4	10.3 / 36.4	78.36
2	16	32.0 / 97.0	12.2 / 84.2	11.7 / 69.6	10.6 / 69.6	72.00
2	64	65.3 / 99.2	11.4 / 95.5	33.6 / 90.1	10.8 / 90.1	84.15
2	256	14.8 / 99.8	11.2 / 98.8	19.8 / 97.3	11.1 / 97.3	87.67
2	1024	11.8 / 100.0	11.1 / 99.7	12.5 / 99.3	11.1 / 99.3	88.58
2	4096	11.3 / 100.0	11.1 / 99.9	11.4 / 99.8	11.1 / 99.8	88.81
4	1	100.0 / 80.0	25.0 / 30.8	9.3 / 19.0	9.8 / 19.0	20.00
4	4	12.9 / 94.1	14.2 / 72.7	10.6 / 53.3	10.7 / 53.3	81.21
4	16	37.2 / 98.5	11.6 / 91.4	21.0 / 82.1	10.8 / 82.1	79.79
4	64	70.3 / 99.6	11.2 / 97.3	50.3 / 94.8	10.8 / 94.8	86.12
4	256	12.6 / 99.7	11.1 / 99.4	20.7 / 98.7	11.1 / 98.7	88.27
4	1024	11.5 / 99.9	11.1 / 99.9	12.6 / 99.7	11.1 / 99.7	88.73
4	4096	11.2 / 100.0	11.1 / 100.0	11.4 / 99.9	11.1 / 99.9	88.85
8	1	4.0 / 88.9	16.0 / 38.1	10.1 / 27.6	10.4 / 27.6	84.85

Ch	Beats/ch	Sink AXIS- in % (byte / beats)	Sink AXI4- wr %	Source AXI4-rd %	Source AXIS-out %	Max abs delta (pp)
8	4	14.5 / 97.0	12.5 / 84.2	11.3 / 69.6	10.6 / 69.6	82.42
8	16	40.5 / 99.2	11.4 / 95.5	34.5 / 90.1	10.8 / 90.1	84.15
8	64	73.1 / 99.8	11.2 / 98.5	66.2 / 97.3	11.0 / 97.3	87.30
8	256	11.8 / 96.1	11.1 / 99.7	21.4 / 99.3	11.1 / 99.3	88.58
8	1024	11.3 / 99.0	11.1 / 99.9	12.6 / 99.8	11.1 / 99.8	88.81
8	4096	11.2 / 99.7	11.1 / 100.0	11.5 / 100.0	11.1 / 100.0	88.87

Result on this build: CHANGED. 112 of 112 beat-aligned cells were compared; 108 differ by more than 0.5 pp; the largest difference is -88.87 pp (ch 8, 4096 beats, sink wr). Cells over the threshold: ch1 b1 sink wr -4.29 pp; ch1 b4 sink sin +20.00 pp; ch1 b4 sink wr -15.00 pp; ch1 b4 source rd -12.92 pp; ch1 b4 source sout -12.70 pp; ch1 b16 sink sin -69.12 pp; ch1 b16 sink wr -59.17 pp; ch1 b16 source rd -42.74 pp; ch1 b16 source sout -42.67 pp; ch1 b64 sink sin -41.32 pp; ch1 b64 sink wr -79.79 pp; ch1 b64 source rd -61.34 pp;

What the byte-checker rows can and cannot show. The byte build's on-chip checkers (BYTE_CRC=1 on axi4_slave_wr_crc_check and axis4_slave_pattern_check) fold the strobed bytes of each beat into the CRC four bytes per cycle and hold ready low while they do, so a 32-byte beat occupies 9 cycles. That caps the harness at $3200 / 9 = 355.6$ MB/s (11.1 % of the 3200 MB/s peak), which is where the large-transfer rows sit. The RAPIDS Beats reference used the word-wide checkers and is not checker-bound. The differences in this table are therefore the checkers backpressuring the DUT, not evidence about the DUT's own throughput, and this table is **not comparable** to the RAPIDS Beats report. The "utilization unchanged" criterion is settled in section 3.1 on the word-wide checker build (BYTE_CRC=0, BUILD.WORD_CRC = 1), where every checker takes one beat per cycle as in the RAPIDS Beats build.

Figure 3.1: Byte minus beats utilization, beat-aligned cells



utilization delta

Beat-aligned rows: bytes, beats, efficiency, MB/s and share of the 3200 MB/s peak

Ch	Beats /ch	Sink bytes	Sink beats	Sink eff	Sink MB/s	Sink % peak	Src bytes	Src beats	Src eff	Src MB/s	Src % peak
1	1	32	1	1.00 0	320	10.0 %	32	1	1.00 0	200	6.2 %
1	4	128	4	1.00 0	800	25.0 %	128	4	1.00 0	298	9.3 %
1	16	512	16	1.00 0	430	13.4 %	512	16	1.00 0	339	10.6 %
1	64	2048	64	1.00 0	372	11.6 %	1056	33	1.00 0	342	10.7 %
1	256	8192	256	1.00 0	359	11.2 %	4864	152	1.00 0	353	11.0 %
1	1024	32768	1024	1.00 0	357	11.1 %	29440	920	1.00 0	355	11.1 %
1	4096	131072	4096	1.00 0	356	11.1 %	127744	3992	1.00 0	355	11.1 %
2	1	64	2	1.00 0	533	16.7 %	64	2	1.00 0	256	8.0 %
2	4	256	8	1.00 0	337	10.5 %	256	8	1.00 0	324	10.1 %
2	16	1024	32	1.00 0	389	12.2 %	928	29	1.00 0	340	10.6 %
2	64	4096	128	1.00 0	363	11.4 %	1312	41	1.00 0	344	10.8 %
2	256	16384	512	1.00 0	357	11.2 %	9152	286	1.00 0	354	11.1 %
2	1024	65536	2048	1.00 0	356	11.1 %	58304	1822	1.00 0	355	11.1 %
2	4096	262144	8192	1.00 0	356	11.1 %	254912	7966	1.00 0	356	11.1 %
4	1	128	4	1.00 0	800	25.0 %	128	4	1.00 0	298	9.3 %

Ch	Beats /ch	Sink bytes	Sink beats	Sink eff	Sink MB/s	Sink % peak	Src bytes	Src beats	Src eff	Src MB/s	Src % peak
4	4	512	16	1.00 0	413	12.9 %	512	16	1.00 0	339	10.6 %
4	16	2048	64	1.00 0	372	11.6 %	1056	33	1.00 0	346	10.8 %
4	64	8192	256	1.00 0	358	11.2 %	1760	55	1.00 0	346	10.8 %
4	256	3276 8	1024	1.00 0	357	11.1 %	1753 6	548	1.00 0	355	11.1 %
4	1024	1310 72	4096	1.00 0	356	11.1 %	1158 40	3620	1.00 0	355	11.1 %
4	4096	5242 88	1638 4	1.00 0	356	11.1 %	5090 56	1590 8	1.00 0	356	11.1 %
8	1	256	8	1.00 0	129	4.0 %	256	8	1.00 0	324	10.1 %
8	4	1024	32	1.00 0	397	12.4 %	960	30	1.00 0	338	10.6 %
8	16	4096	128	1.00 0	363	11.4 %	1280	40	1.00 0	345	10.8 %
8	64	1638 4	512	1.00 0	357	11.1 %	2720	85	1.00 0	352	11.0 %
8	256	6553 6	2048	1.00 0	356	11.1 %	3392 0	1060	1.00 0	355	11.1 %
8	1024	2621 44	8192	1.00 0	356	11.1 %	2305 28	7204	1.00 0	356	11.1 %
8	4096	1048 576	3276 8	1.00 0	356	11.1 %	1016 960	3178 0	1.00 0	356	11.1 %

Every beat-aligned row moves whole beats, so efficiency is 1.000 by construction; measured range over 56 rows: 1.000 to 1.000.

4. Transfer size (offset 0)

One descriptor per channel, payload in bytes, start offset 0. Bytes moved and beats are totals over the active channels; efficiency is payload bytes over beats times 32 bytes per beat, on the AXIS

side and on the memory side. MB/s is total bytes over the measurement window (the longer of the stream and memory windows) and is always shown with its share of the 3200 MB/s peak (100 MHz x 32 B).

4.1 Sink path

Size sweep, sink path, offset 0

Payload B	Offset	Ch	Bytes moved	AXIS beats	Mem beats	Eff (AXIS)	Eff (mem)	MB/s	% of peak	Result
1	0	1	1	1	1	0.031	0.031	10.0	0.31 %	PASS
1	0	2	2	2	2	0.031	0.031	16.7	0.52 %	PASS
1	0	4	4	4	4	0.031	0.031	25.0	0.78 %	PASS
1	0	8	8	8	8	0.031	0.031	4.0	0.13 %	PASS
7	0	1	7	1	1	0.219	0.219	70.0	2.19 %	PASS
7	0	2	14	2	2	0.219	0.219	116.7	3.65 %	PASS
7	0	4	28	4	4	0.219	0.219	175.0	5.47 %	PASS
7	0	8	56	8	8	0.219	0.219	28.1	0.88 %	PASS
32	0	1	32	1	1	1.000	1.000	320.0	10.00 %	PASS
32	0	2	64	2	2	1.000	1.000	533.3	16.67 %	PASS
32	0	4	128	4	4	1.000	1.000	800.0	25.00 %	PASS
32	0	8	256	8	8	1.000	1.000	128.6	4.02 %	PASS
33	0	1	33	2	2	0.516	0.516	275.0	8.59 %	PASS

Payload B	Offset	Ch	Bytes moved	AXIS beats	Mem beats	Eff (AXIS)	Eff (mem)	MB/s	% of peak	Result
33	0	2	66	4	4	0.516	0.516	471.4	14.73 %	PASS
33	0	4	132	8	8	0.516	0.516	113.8	3.56 %	PASS
33	0	8	264	16	16	0.516	0.516	129.4	4.04 %	PASS
64	0	1	64	2	2	1.000	1.000	533.3	16.67 %	PASS
64	0	2	128	4	4	1.000	1.000	914.3	28.57 %	PASS
64	0	4	256	8	8	1.000	1.000	220.7	6.90 %	PASS
64	0	8	512	16	16	1.000	1.000	251.0	7.84 %	PASS
77	0	1	77	3	3	0.802	0.802	550.0	17.19 %	PASS
77	0	2	154	6	6	0.802	0.802	208.1	6.50 %	PASS
77	0	4	308	12	12	0.802	0.802	256.7	8.02 %	PASS
77	0	8	616	24	24	0.802	0.802	290.6	9.08 %	PASS
203	0	1	203	7	7	0.906	0.906	369.1	11.53 %	PASS
203	0	2	406	14	14	0.906	0.906	431.9	13.50 %	PASS
203	0	4	812	28	28	0.906	0.906	386.7	12.08 %	PASS
203	0	8	1624	56	56	0.906	0.906	367.4	11.48 %	PASS
1024	0	1	1024	32	32	1.000	1.000	389.4	12.17 %	PASS

Payload B	Offset	Ch	Bytes moved	AXIS beats	Mem beats	Eff (AXIS)	Eff (mem)	MB/s	% of peak	Result
1024	0	2	2048	64	64	1.000	1.000	371.7	11.62 %	PASS
1024	0	4	4096	128	128	1.000	1.000	363.4	11.36 %	PASS
1024	0	8	8192	256	256	1.000	1.000	359.5	11.23 %	PASS
4035	0	1	4035	127	127	0.993	0.993	360.9	11.28 %	PASS
4035	0	2	8070	254	254	0.993	0.993	358.0	11.19 %	PASS
4035	0	4	16140	508	508	0.993	0.993	356.1	11.13 %	PASS
4035	0	8	32280	1016	1016	0.993	0.993	355.6	11.11 %	PASS
4096	0	1	4096	128	128	1.000	1.000	363.4	11.36 %	PASS
4096	0	2	8192	256	256	1.000	1.000	359.5	11.23 %	PASS
4096	0	4	16384	512	512	1.000	1.000	357.5	11.17 %	PASS
4096	0	8	32768	1024	1024	1.000	1.000	356.5	11.14 %	PASS

4.2 Source path

Size sweep, source path, offset 0

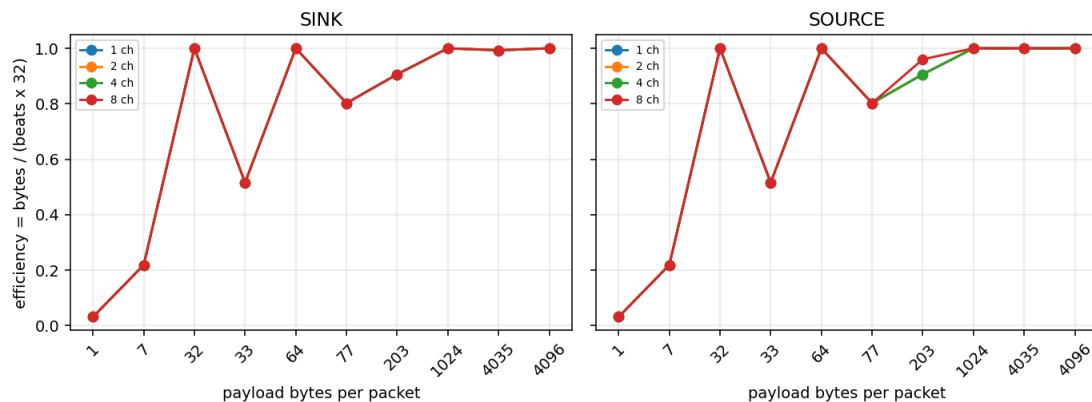
Payload B	Offset	Ch	Bytes moved	AXIS beats	Mem beats	Eff (AXIS)	Eff (mem)	MB/s	% of peak	Result
1	0	1	1	1	1	0.031	0.031	6.2	0.20 %	PASS
1	0	2	2	2	2	0.031	0.031	11.1	0.35 %	PASS

Payl oad B	Offset	Ch	Bytes move d	AXIS beats	Mem beats	Eff (AXIS)	Eff (mem)	MB/s	% of peak	Result
1	0	4	4	4	4	0.031	0.031	18.2	0.57 %	PASS
1	0	8	8	8	8	0.031	0.031	26.7	0.83 %	PASS
7	0	1	7	1	1	0.219	0.219	43.8	1.37 %	PASS
7	0	2	14	2	2	0.219	0.219	73.7	2.30 %	PASS
7	0	4	28	4	4	0.219	0.219	112.0	3.50 %	PASS
7	0	8	56	8	8	0.219	0.219	151.4	4.73 %	PASS
32	0	1	32	1	1	1.000	1.000	200.0	6.25 %	PASS
32	0	2	64	2	2	1.000	1.000	256.0	8.00 %	PASS
32	0	4	128	4	4	1.000	1.000	297.7	9.30 %	PASS
32	0	8	256	8	8	1.000	1.000	324.1	10.13 %	PASS
33	0	1	33	2	2	0.516	0.516	132.0	4.12 %	PASS
33	0	2	66	4	4	0.516	0.516	183.3	5.73 %	PASS
33	0	4	132	8	8	0.516	0.516	227.6	7.11 %	PASS
33	0	8	264	16	16	0.516	0.516	258.8	8.09 %	PASS
64	0	1	64	2	2	1.000	1.000	256.0	8.00 %	PASS
64	0	2	128	4	4	1.000	1.000	297.7	9.30 %	PASS

Payload B	Offset	Ch	Bytes moved	AXIS beats	Mem beats	Eff (AXIS)	Eff (mem)	MB/s	% of peak	Result
64	0	4	256	8	8	1.000	1.000	324.1	10.13 %	PASS
64	0	8	512	16	16	1.000	1.000	339.1	10.60 %	PASS
77	0	1	77	3	3	0.802	0.802	226.5	7.08 %	PASS
77	0	2	154	6	6	0.802	0.802	270.2	8.44 %	PASS
77	0	4	308	12	12	0.802	0.802	299.0	9.34 %	PASS
77	0	8	616	24	24	0.802	0.802	315.9	9.87 %	PASS
203	0	1	203	7	7	0.906	0.906	290.0	9.06 %	PASS
203	0	2	406	14	14	0.906	0.906	317.2	9.91 %	PASS
203	0	4	812	28	28	0.906	0.906	332.8	10.40 %	PASS
203	0	8	1014	33	56	0.960	0.566	339.1	10.60 %	PASS
1024	0	1	928	29	32	1.000	0.906	339.9	10.62 %	PASS
1024	0	2	1056	33	64	1.000	0.516	341.7	10.68 %	PASS
1024	0	4	1280	40	128	1.000	0.312	343.2	10.72 %	PASS
1024	0	8	1760	55	256	1.000	0.215	348.5	10.89 %	PASS
4035	0	1	1280	40	127	1.000	0.315	344.1	10.75 %	PASS
4035	0	2	1760	55	254	1.000	0.217	347.1	10.85 %	PASS

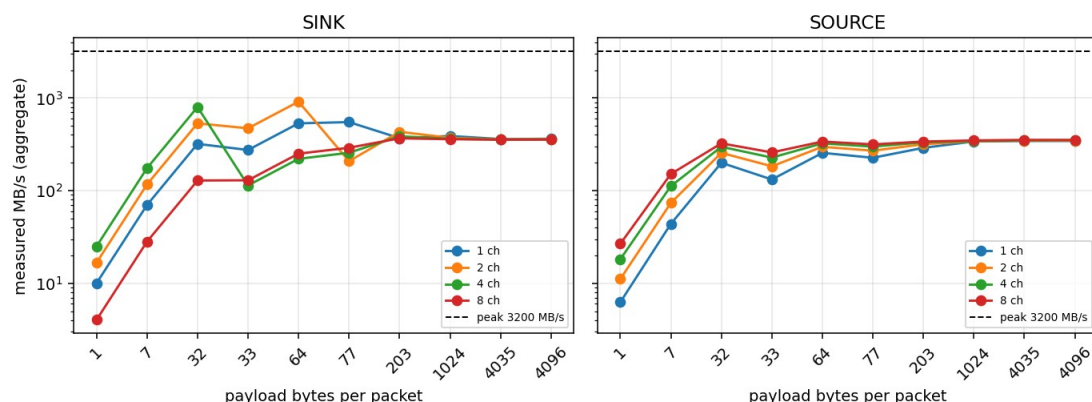
Payl oad B	Offset	Ch	Bytes move d	AXIS beats	Mem beats	Eff (AXIS)	Eff (mem)	MB/s	% of peak	Result
4035	0	4	2656	83	508	1.000	0.163	349.0	10.91 %	PASS
4035	0	8	4512	141	1016	1.000	0.139	353.3	11.04 %	PASS
4096	0	1	1280	40	128	1.000	0.312	344.1	10.75 %	PASS
4096	0	2	1760	55	256	1.000	0.215	347.1	10.85 %	PASS
4096	0	4	2688	84	512	1.000	0.164	352.3	11.01 %	PASS
4096	0	8	4512	141	1024	1.000	0.138	352.0	11.00 %	PASS

Figure 4.1: Efficiency against payload size



efficiency vs size

Figure 4.2: Measured MB/s against payload size, peak as the dashed line



MB/s vs size

Observations from this data:

- Sink, 8 channels, 4096 B: 357 MB/s = 11.1 % of the 3200 MB/s peak at efficiency 1.000.
- Sink, 8 channels: 1 B payload reaches 4.0 MB/s, 89x less than 4096 B; the window is dominated by fixed per-packet latency (199 cycles for one beat per channel).
- Source, 8 channels, 4096 B: 352 MB/s = 11.0 % of the 3200 MB/s peak at efficiency 1.000.
- Source, 8 channels: 1 B payload reaches 26.7 MB/s, 13x less than 4096 B; the window is dominated by fixed per-packet latency (30 cycles for one beat per channel).

5. Start offset

A descriptor may start mid-beat. The offset moves payload into an extra memory beat when offset + payload crosses a beat boundary, lowering memory-side efficiency while the AXIS side, which packs from lane 0, is unchanged. Offset 0 rows repeat the size sweep.

5.1 Sink path

Offset sweep, sink path

Payl oad B	Offset	Ch	Bytes move d	AXIS beats	Mem beats	Eff (AXIS)	Eff (mem)	MB/s	% of peak	Result
33	0	1	33	2	2	0.516	0.516	275.0	8.59 %	PASS

Payload B	Offset	Ch	Bytes moved	AXIS beats	Mem beats	Eff (AXIS)	Eff (mem)	MB/s	% of peak	Result
33	1	1	33	2	2	0.516	0.516	275.0	8.59 %	PASS
33	5	1	33	2	2	0.516	0.516	275.0	8.59 %	PASS
33	31	1	33	2	2	0.516	0.516	275.0	8.59 %	PASS
203	0	1	203	7	7	0.906	0.906	369.1	11.53 %	PASS
203	1	1	203	7	7	0.906	0.906	369.1	11.53 %	PASS
203	5	1	203	7	7	0.906	0.906	369.1	11.53 %	PASS
203	31	1	203	7	8	0.906	0.793	225.6	7.05 %	PASS
1024	0	1	1024	32	32	1.000	1.000	389.4	12.17 %	PASS
1024	1	1	1024	32	33	1.000	0.970	317.0	9.91 %	PASS
1024	5	1	1024	32	33	1.000	0.970	318.0	9.94 %	PASS
1024	31	1	1024	32	33	1.000	0.970	324.1	10.13 %	PASS
4035	0	1	4035	127	127	0.993	0.993	360.9	11.28 %	PASS
4035	1	1	4035	127	127	0.993	0.993	360.9	11.28 %	PASS
4035	5	1	4035	127	127	0.993	0.993	361.2	11.29 %	PASS
4035	31	1	4035	127	128	0.993	0.985	344.6	10.77 %	PASS
33	0	8	264	16	16	0.516	0.516	129.4	4.04 %	PASS

Payl oad B	Offset	Ch	Bytes move d	AXIS beats	Mem beats	Eff (AXIS)	Eff (mem)	MB/s	% of peak	Result
33	1	8	264	16	16	0.516	0.516	129.4	4.04 %	PASS
33	5	8	264	16	16	0.516	0.516	129.4	4.04 %	PASS
33	31	8	264	16	16	0.516	0.516	129.4	4.04 %	PASS
203	0	8	1624	56	56	0.906	0.906	367.4	11.48 %	PASS
203	1	8	1624	56	56	0.906	0.906	367.4	11.48 %	PASS
203	5	8	1624	56	56	0.906	0.906	368.3	11.51 %	PASS
203	31	8	1624	56	64	0.906	0.793	249.8	7.81 %	PASS
102 4	0	8	8192	256	256	1.000	1.000	359.5	11.23 %	PASS
102 4	1	8	8192	256	264	1.000	0.970	328.6	10.27 %	PASS
102 4	5	8	8192	256	264	1.000	0.970	328.7	10.27 %	PASS
102 4	31	8	8192	256	264	1.000	0.970	329.5	10.30 %	PASS
403 5	0	8	3228 0	1016	1016	0.993	0.993	355.6	11.11 %	PASS
403 5	1	8	3228 0	1016	1016	0.993	0.993	355.6	11.11 %	PASS
403 5	5	8	3228 0	1016	1016	0.993	0.993	355.7	11.12 %	PASS
403 5	31	8	3228 0	1016	1024	0.993	0.985	347.4	10.86 %	PASS

5.2 Source path

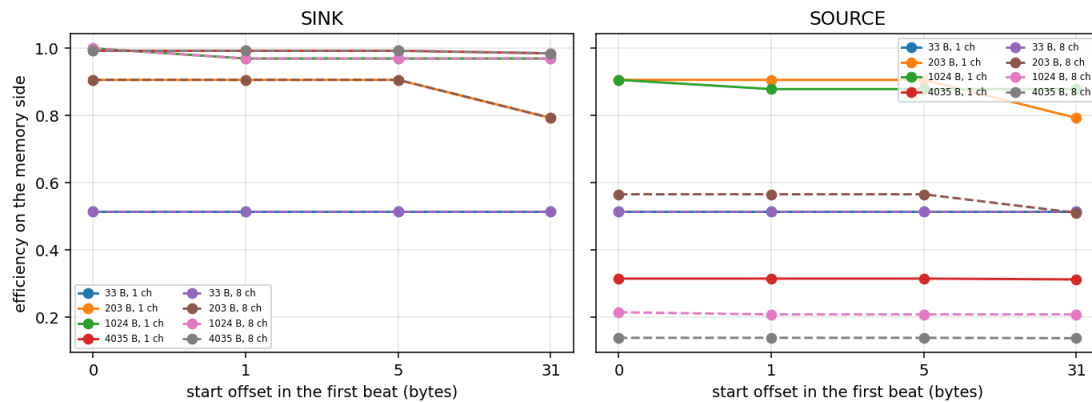
Offset sweep, source path

Payload B	Offset	Ch	Bytes moved	AXIS beats	Mem beats	Eff (AXIS)	Eff (mem)	MB/s	% of peak	Result
33	0	1	33	2	2	0.516	0.516	132.0	4.12 %	PASS
33	1	1	33	2	2	0.516	0.516	126.9	3.97 %	PASS
33	5	1	33	2	2	0.516	0.516	126.9	3.97 %	PASS
33	31	1	33	2	2	0.516	0.516	126.9	3.97 %	PASS
203	0	1	203	7	7	0.906	0.906	290.0	9.06 %	PASS
203	1	1	203	7	7	0.906	0.906	285.9	8.93 %	PASS
203	5	1	203	7	7	0.906	0.906	285.9	8.93 %	PASS
203	31	1	203	7	8	0.906	0.793	285.9	8.93 %	PASS
1024	0	1	928	29	32	1.000	0.906	339.9	10.62 %	PASS
1024	1	1	928	29	33	1.000	0.879	339.9	10.62 %	PASS
1024	5	1	928	29	33	1.000	0.879	339.9	10.62 %	PASS
1024	31	1	928	29	33	1.000	0.879	339.9	10.62 %	PASS
4035	0	1	1280	40	127	1.000	0.315	344.1	10.75 %	PASS
4035	1	1	1280	40	127	1.000	0.315	344.1	10.75 %	PASS
4035	5	1	1280	40	127	1.000	0.315	344.1	10.75 %	PASS

Payload B	Offset	Ch	Bytes moved	AXIS beats	Mem beats	Eff (AXIS)	Eff (mem)	MB/s	% of peak	Result
5									%	
4035	31	1	1280	40	128	1.000	0.312	344.1	10.75	PASS
33	0	8	264	16	16	0.516	0.516	258.8	8.09	PASS
33	1	8	264	16	16	0.516	0.516	256.3	8.01	PASS
33	5	8	264	16	16	0.516	0.516	256.3	8.01	PASS
33	31	8	264	16	16	0.516	0.516	256.3	8.01	PASS
203	0	8	1014	33	56	0.960	0.566	339.1	10.60	PASS
203	1	8	1014	33	56	0.960	0.566	339.1	10.60	PASS
203	5	8	1014	33	56	0.960	0.566	339.1	10.60	PASS
203	31	8	1046	34	64	0.961	0.511	344.1	10.75	PASS
1024	0	8	1760	55	256	1.000	0.215	348.5	10.89	PASS
1024	1	8	1760	55	264	1.000	0.208	345.1	10.78	PASS
1024	5	8	1760	55	264	1.000	0.208	345.1	10.78	PASS
1024	31	8	1760	55	264	1.000	0.208	345.1	10.78	PASS
4035	0	8	4512	141	1016	1.000	0.139	353.3	11.04	PASS
4035	1	8	4512	141	1016	1.000	0.139	353.3	11.04	PASS

Payl oad B	Offset	Ch	Bytes move d	AXIS beats	Mem beats	Eff (AXIS)	Eff (mem)	MB/s	% of peak	Result
403 5	5	8	4512	141	1016	1.000	0.139	353.3	11.04 %	PASS
403 5	31	8	4512	141	1024	1.000	0.138	352.0	11.00 %	PASS

Figure 5.1: Memory-side efficiency against start offset



efficiency vs offset

Offsets that cost an extra memory beat (sink, 1 channel): 203 B at offset 31: 7 -> 8 memory beats; 1024 B at offset 1: 32 -> 33 memory beats; 1024 B at offset 5: 32 -> 33 memory beats; 1024 B at offset 31: 32 -> 33 memory beats; 4035 B at offset 31: 127 -> 128 memory beats.

6. Descriptor chains

Several descriptors per channel back to back (4 channels), offset 0. Efficiency counts payload over all beats of the chain.

Descriptor chains, 4 channels

Paylo ad B	Descs	Path	Bytes moved	AXIS beats	Eff (AXIS)	MB/s	% of peak	Result
1	16	sink	64	64	0.031	5.4	0.17 %	PASS
1	16	source	64	64	0.031	37.2	1.16 %	PASS
64	4	sink	1024	32	1.000	223.6	6.99 %	PASS
64	4	source	1024	32	1.000	347.1	10.85 %	PASS

Paylo ad B	Descs	Path	Bytes moved	AXIS beats	Eff (AXIS)	MB/s	% of peak	Result
64	16	sink	4096	128	1.000	221.4	6.92 %	PASS
64	16	source	3872	121	1.000	352.6	11.02 %	PASS
1024	4	sink	16384	512	1.000	341.4	10.67 %	PASS
1024	4	source	2656	83	1.000	350.9	10.96 %	PASS
1024	16	sink	65536	2048	1.000	336.3	10.51 %	PASS
1024	16	source	50432	1576	1.000	355.4	11.11 %	PASS

7. Source backpressure (integrity only)

The source backpressure is paced by the host toggling the checker's ready over UART, so the cycle counts measure the host, not the DUT (rapids TASK-085). These rows are integrity checks: bytes and beats must still match and the golden CRCs must agree. Their MB/s is not reported.

Source with backpressure, integrity rows

Ch	Payload B	Offset	Expected bytes	Expected AXIS beats	MB/s	Result
1	33	0	33	2	n/a (host-paced)	PASS
1	33	5	33	2	n/a (host-paced)	PASS
1	1024	0	1024	32	n/a (host-paced)	PASS
1	1024	5	1024	32	n/a (host-paced)	PASS
4	33	0	132	8	n/a (host-paced)	PASS
4	33	5	132	8	n/a (host-paced)	PASS
4	1024	0	4096	128	n/a (host-paced)	PASS

Ch	Payload B	Offset	Expected bytes	Expected AXIS beats	MB/s	Result
4	1024	5	4096	128	n/a (host-paced)	PASS

8. Failures, caveats and coverage

No measured point failed.

Coverage: 117 of 117 planned points of profile standard are in this file (117 passed, 0 failed).

Standing limitations of the design, not of this measurement:

- The sink `s_axis_tready` is one signal qualified by TID, so a beat for a channel whose packet record has not arrived blocks every channel behind it on the stream (head-of-line blocking, inherent and documented).
- TYPE=EXT descriptors stay beat-aligned by design and are not part of the byte sweeps.
- AXI error responses (RRESP and BRESP other than OKAY) are not exercised by the board campaign or by the directed sequences: the harness memory model always answers OKAY and has no fault-injection hook, and no RAPIDS test in the tree drives a non-OKAY response on the read or write master. The response-error flag path is therefore unproven on silicon and in simulation. Closing this needs an injection hook in the harness, which is not built.
- Each interface has its own measurement window; the `sin` window runs from the first to the last stream beat (rapids ISSUE-001), so windows differ per interface and MB/s here uses the longer of the stream and memory windows.

9. Build and resources

Post-route resources and timing at 100 MHz, XC7K325T (203,800 LUTs, 445 BRAM tiles)

Build	Slice LUTs	Slice regs	BRAM tiles	WNS setup	WHS hold	Failing endpoint s	Bitstream sha256
bytecrc	92,874	82,379	68	+0.301 ns	+0.023 ns	0	cfad34c3465f11

Build	Slice LUTs	Slice regs	BRAM tiles	WNS setup	WHS hold	Failing endpoint s	Bitstream sha256
							10
word	86,065	79,527	52	+0.077	+0.013	0	48e1282
crc				ns	ns		c46e707
							cd

Build configuration of each row:

- **bytecrc:** BYTE_CRC=1 USE_OBSERVERS=1 OBS_ENABLE_MON_TAPS=0 (channel-reset + hold-junk fixes).
- **wordcrc:** BYTE_CRC=0 USE_OBSERVERS=1 OBS_ENABLE_MON_TAPS=0. RTL commit d1d20b046.

Every build closes timing at 100 MHz with no failing endpoint. The slack is small and positive; treat 100 MHz as the design point, not as margin. The resource figures come from the post-route utilization report; the word-wide checker build is a measurement bitstream and is not the standard one.

10. Provenance and reproduction

Provenance

Item	Value
Results file	rapids_byte_perf_20260930_142731.json
Timestamp	2026-09-30T14:27:36
Profile	standard
Status	final
Bitstream	rapids_byte.bit sha256 cfad34c3465f1110
Design	256-bit, 8 channels, 4096 B SRAM per channel, 100 MHz, peak 3200 MB/s per direction
Beats reference	genesys_dw256_obs_C.json

Device readback per session

Session	Bitstream sha256	CSR_ID	BUILD	Sentinel start	Sentinel end	Stable	Aborted
1	cfad34c3465f1110	0x52415042	0x0A070820	0x00000000	0x00000000	yes	no

The sentinel is the MON_LIMIT register written once at configure time. It resets on any FPGA reconfiguration, so a change between readbacks means the device was reprogrammed during the run. A point measured across such a change is dropped, not recorded. Device stable for the whole run: yes.

One command runs the campaign and regenerates this report:

```
cd projects/fpga-systems/Genesys2/rapids/flows-rapids
./byte_perf.sh --profile full --final          # final numbers (standard
byte-CRC bitstream)
./byte_perf.sh --profile standard              # quick run, writes
*_prelim_*.json
```

The word-wide aligned results are rapids_byte_aligned_wordcrc_20260930.json; its device readback follows.

Device readback per session

Session	Bitstream sha256	CSR_ID	BUILD	Sentinel start	Sentinel end	Stable	Aborted
1	48e1282c46e707cd	0x52415042	0x1A070820	0x00000000	0x00000000	yes	no

The sentinel is the MON_LIMIT register written once at configure time. It resets on any FPGA reconfiguration, so a change between readbacks means the device was reprogrammed during the run. A point measured across such a change is dropped, not recorded. Device stable for the whole run: yes.